Decision Trees

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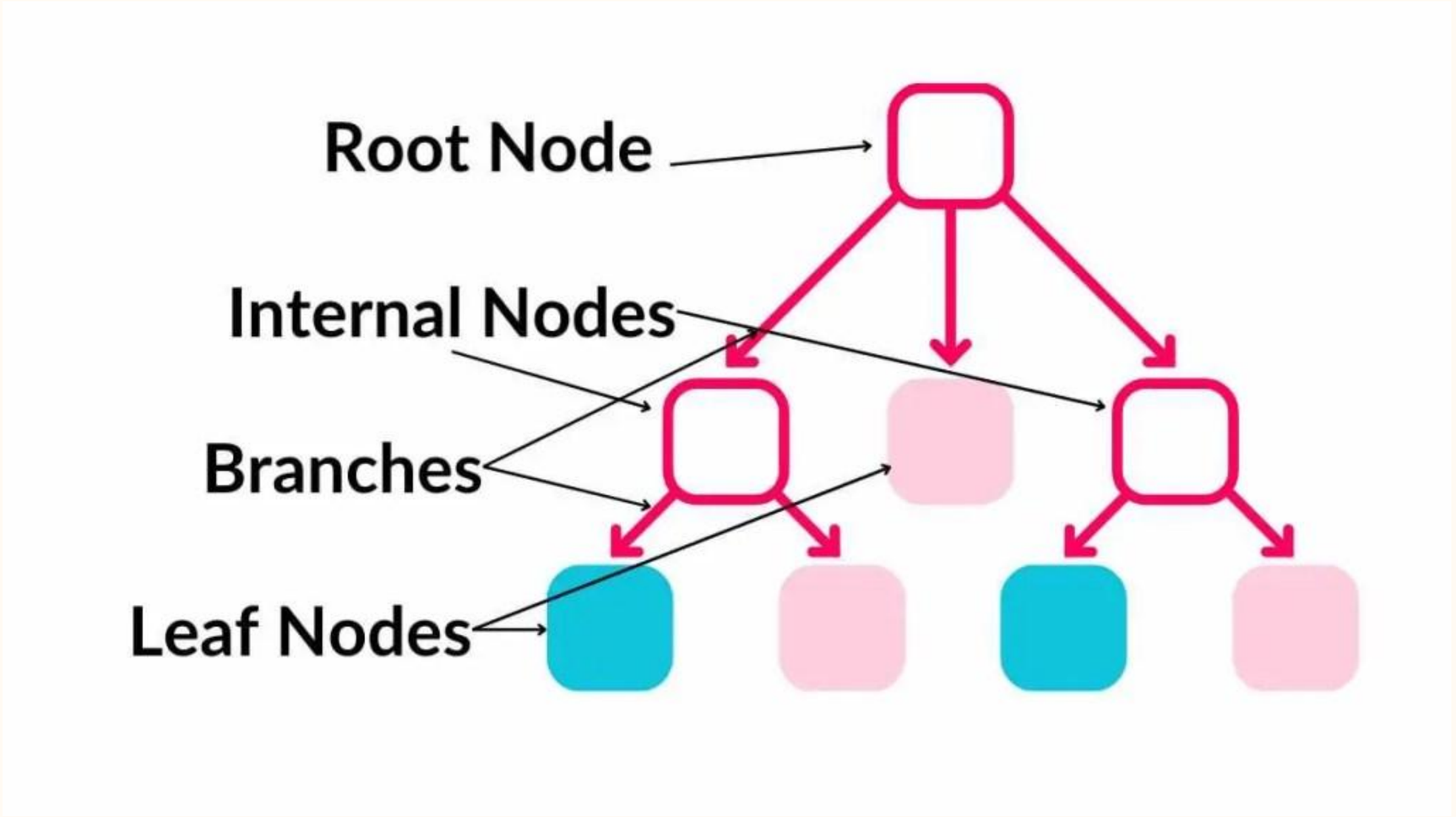
Introduction to Decision Trees

Definition

- A Decision Tree is a supervised machine learning algorithm used for classification and regression tasks.
- It makes predictions by learning a sequence of decision rules from training data.
- Each internal node represents a test on a feature.
- Each branch represents the outcome of a test.
- Each leaf node represents the final prediction.

Key Characteristics

- Easy to understand and interpret.
- Can model nonlinear relationships.
- Requires little feature scaling compared with many other algorithms.
- Can be used for both categorical and numerical target variables.


Root Node
Internal Nodes
Branches
Leaf Nodes

Structure of a Decision Tree

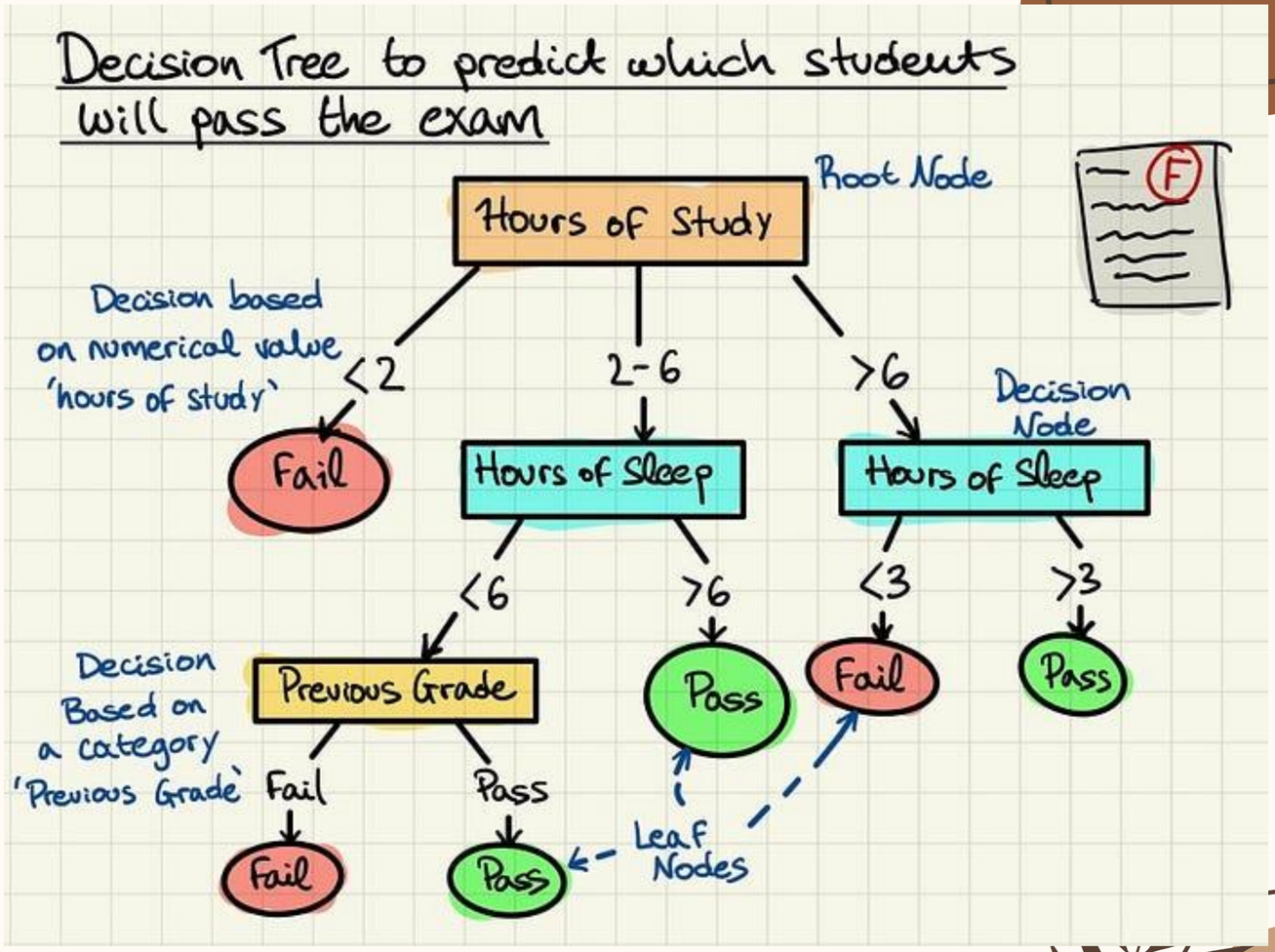
Main Components

1. Root Node
    - The starting point of the tree.
    - Represents the first decision or feature test.
2. Decision Node (Internal Node)
    - Represents a condition applied to a feature.
    - Divides the data into smaller groups.
3. Branch
    - Represents the outcome of a decision.
    - Connects a parent node to a child node.
4. Leaf Node
    - Represents the final prediction.
    - In classification, it predicts a class.
    - In regression, it predicts a numerical value.

Structure of a Decision Tree

Prediction Process

1. Start at the root node.
2. Evaluate the condition at the current node.
3. Follow the corresponding branch.
4. Continue until a leaf node is reached.
5. Return the prediction associated with the leaf.



Types of Decision Trees

1. Decision Tree for Classification

- Used when the target variable represents a category or class.
- Predicts discrete class labels.
- Common examples include spam detection, disease classification, and student pass/fail prediction.
- Common splitting criteria include Gini Impurity and Entropy.

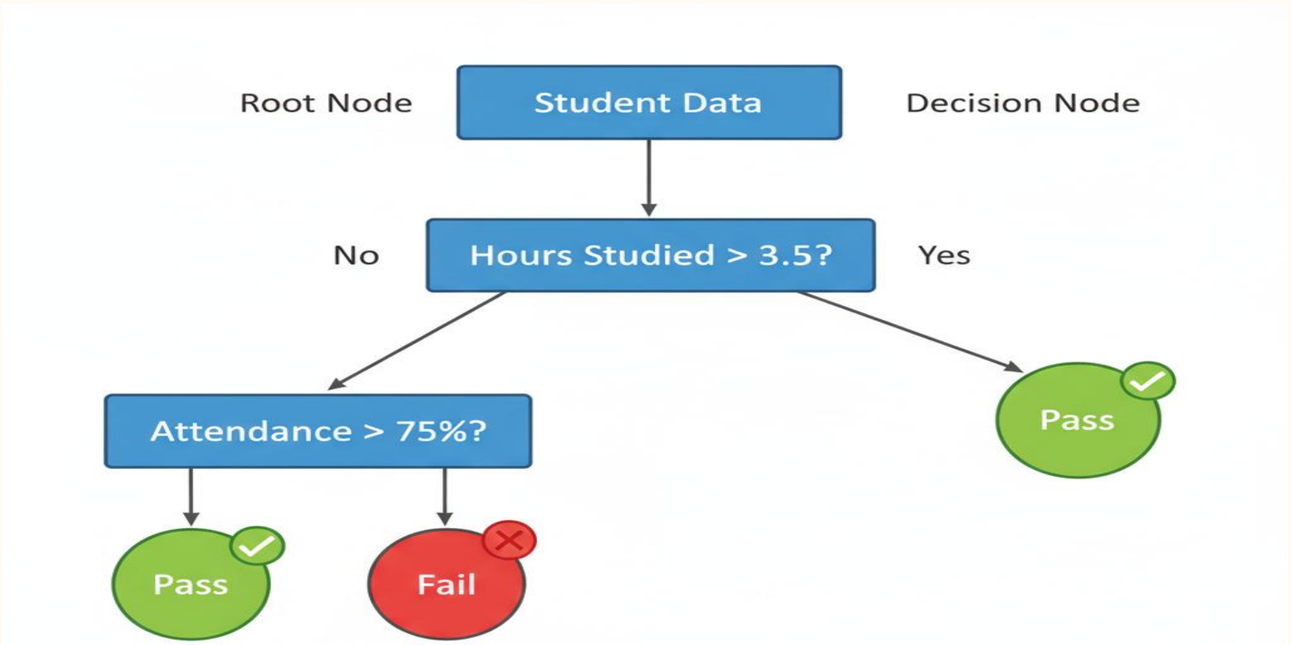
Example: Predicting whether a student will pass or fail.

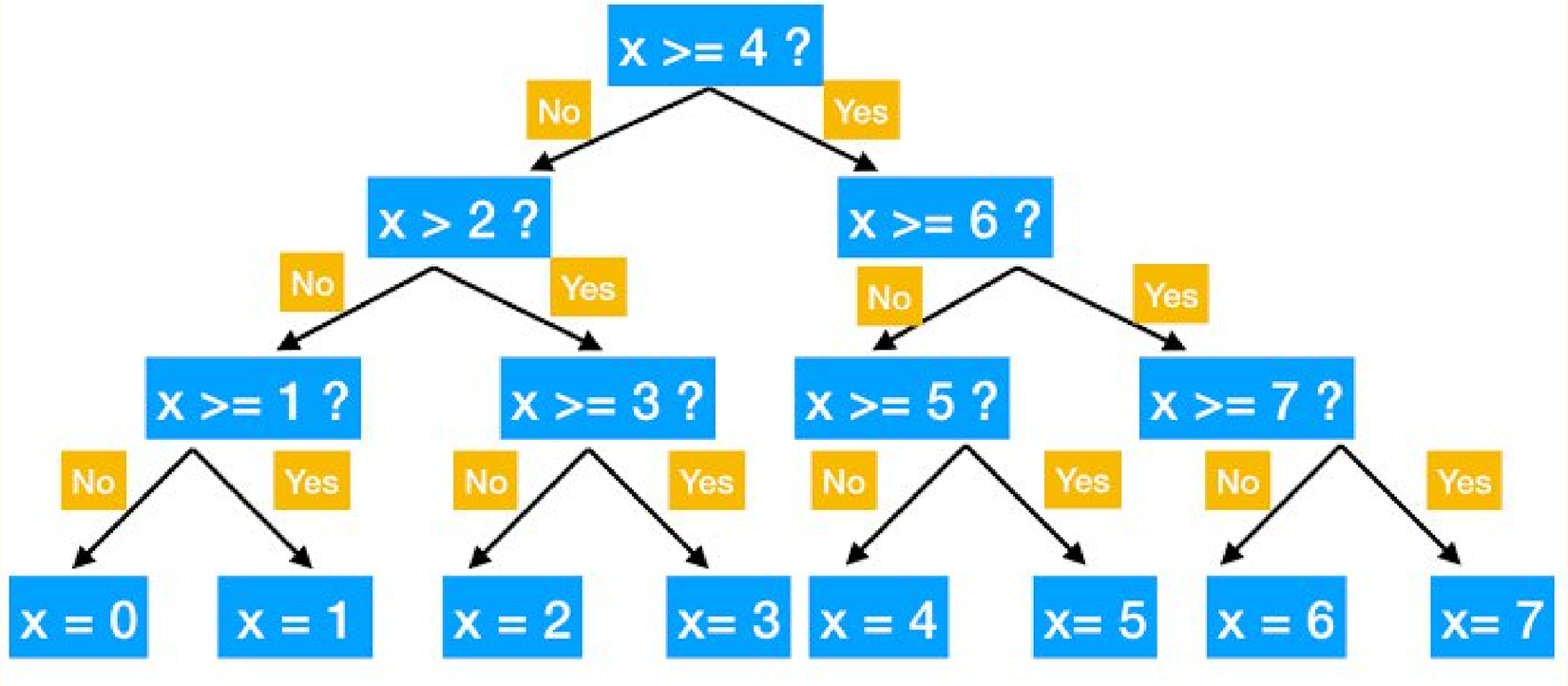
2. Decision Tree for Regression

- Used when the target variable is numerical.
- Predicts a numerical value.
- Common examples include house-price prediction, sales forecasting, and score prediction.
- Splits can be selected using criteria based on variance reduction or squared-error reduction.

Example: Predicting a student's final exam score.

Key Difference: Classification predicts a **class / Regression** predicts a **numerical** value**.**


Root Node
Student Data
Decision Node
No
Hours Studied > 3.5?
Yes
Attendance > 75%?
Pass
Pass
Fail


x >= 4 ?
No
Yes
x > 2 ?
x >= 6 ?
No
Yes
No
Yes
x >= 1 ?
x >= 3 ?
x >= 5 ?
x >= 7 ?
No
Yes
No
Yes
No
Yes
No
Yes
x = 0
x = 1
x = 2
x= 3
x = 4
x= 5
x = 6
x= 7

Learning a Decision Tree and Choosing the Best Split

How Does a Decision Tree Learn?

- Start with the entire training dataset.
- Evaluate candidate features and possible splits.
- Select the split that best improves the chosen criterion.
- Divide the dataset into smaller groups.
- Repeat the process recursively for each group.

What Is a Good Split?

- A good split separates examples into groups with more consistent target values.
- In classification, a pure node contains examples belonging to the same class.
- The algorithm evaluates how much uncertainty is reduced after a split.

Common Splitting Criteria

- **Entropy:** Measures uncertainty in the class distribution.
- **Information Gain:** Measures the reduction in entropy after a split.
- **Gini Impurity:** Measures the impurity of class labels.

Important Principle

For Information Gain, the preferred split is the one with the highest Information Gain.

How Does a Decision Tree Choose a Split?

The Splitting Process

- The algorithm evaluates possible splits on available features.
- A split divides the training data into smaller groups.
- The algorithm selects a split according to the chosen evaluation criterion.
- The process continues recursively until a stopping condition is reached.

Common Splitting Criteria

1. **For Classification**
   - Gini Impurity measures the impurity of class labels within a node.
   - Entropy measures the uncertainty associated with the class distribution.
   - Information Gain measures the reduction in entropy after a split.
2. **For Regression**
   - Variance Reduction measures the reduction in target variability.
   - Squared Error Reduction measures the improvement in prediction error after splitting.

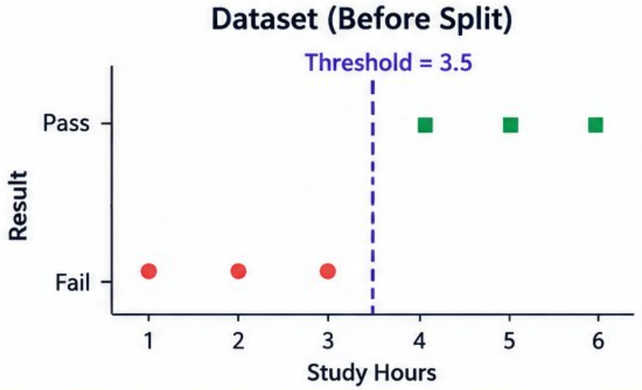
Important Note

The best split depends on the data, the target variable, and the selected criterion.

How Does a Decision Tree Choose a Split?

Original Dataset

ID	Study Hours	Result
A	1	Fail
B	2	Fail
C	3	Fail
D	4	Pass
E	5	Pass
F	6	Pass



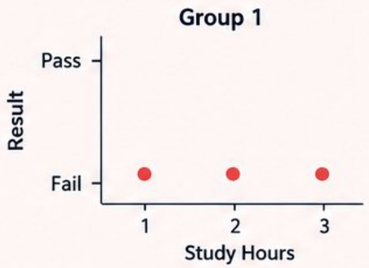
Split on: **Study Hours ≤ 3.5?**

Yes (≤ 3.5)

No (> 3.5)

Group 1 (Study Hours ≤ 3.5)

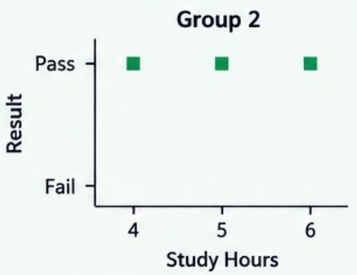
ID	Study Hours	Result
A	1	Fail
B	2	Fail
C	3	Fail



Prediction: Fail (majority class)

Group 2 (Study Hours > 3.5)

ID	Study Hours	Result
D	4	Pass
E	5	Pass
F	6	Pass



Prediction: Pass (majority class)

Entropy — Measuring Uncertainty

Definition: Entropy measures the uncertainty or impurity of the class distribution in a dataset.

For binary classification:

$$H(Y) = -\sum_{y \in Y} p(y) \log_2 p(y)$$

Case 1: Pure Dataset

- 4 Pass, 0 Fail.
- Entropy = 0.
- The class is completely predictable.

Case 2: Balanced Dataset

- 4 Pass, 4 Fail.
- Entropy = 1 bit.
- Uncertainty is maximum for binary classification.

Key Observation

- Lower entropy indicates less uncertainty.
- Higher entropy indicates greater uncertainty.

Information Gain — Selecting the Best Split

Definition: Information Gain measures the reduction in entropy achieved by splitting a dataset using a feature.

$$Gain(S, a) = Entropy(S) - \sum_{v \in values(S)} \frac{|S_v|}{|S|} Entropy(S_v)$$

Calculation Steps

1. Calculate the entropy of the original dataset.
2. Calculate the entropy of each resulting group.
3. Compute the weighted average of the groups' entropies.
4. Subtract the weighted entropy from the original entropy.

Student Pass/Fail Prediction

A dataset contains the study hours and examination results of six students.

Student	Study Hours	Result
A	1	Fail
B	2	Fail
C	3	Fail
D	4	Pass
E	5	Pass
F	6	Pass

How is the threshold selected?

The threshold is not necessarily the mean of the feature values. In a Decision Tree, candidate thresholds are evaluated to find a split that best separates the target classes according to a selected criterion, such as Gini Impurity or Information Gain. In our example, 3.5 is selected because it lies between 3 hours (Fail) and 4 hours (Pass), perfectly separating the two classes.

Student Pass/Fail Prediction

Student	Weather	Temperature	Play Sports?
1	Sunny	Hot	No
2	Sunny	Hot	No
3	Sunny	Mild	No
4	Sunny	Cool	Yes
5	Rainy	Cool	Yes
6	Rainy	Mild	Yes
7	Rainy	Cool	Yes
8	Rainy	Hot	No

The **goal** of a Decision Tree split is to reduce uncertainty in the resulting groups. Lower entropy indicates purer groups, while higher information gain indicates a greater reduction in uncertainty. Therefore, **when using Information Gain, we prefer the split with the highest Information Gain.**

Overfitting, Advantages, and Limitations

Overfitting

- Occurs when a tree learns the training data too closely.
- May result in excellent training performance but poor generalization to unseen data.
- Can be controlled using depth limits, minimum sample requirements, and validation techniques.

Advantages

- Easy to understand and interpret.
- Can model nonlinear relationships.
- Handles classification and regression tasks.
- Requires relatively little feature preprocessing.

Limitations

- Deep trees may overfit the training data.
- Small changes in training data can produce different trees.
- Greedy splitting does not guarantee a globally optimal tree.
- Complex trees can become difficult to interpret.

Conclusion

Decision Trees learn decision rules from data and use these rules to make predictions. Their performance depends on appropriate splitting criteria, model complexity, and evaluation on unseen data.

Q1. What is the difference between a Decision Tree for Classification and a Decision Tree for Regression?

✓ A classification tree predicts categorical classes, whereas a regression tree predicts numerical values.

Q2. What is the purpose of max_depth in a Decision Tree?

✓ It limits the depth of the tree and can help reduce overfitting.

Q3. A Decision Tree predicts whether an email is spam or not spam. Is this classification or regression? Why?

✓ Classification, because the target variable represents categorical classes.

Summary

- Decision Trees learn decision rules from training data.
- Entropy measures uncertainty in classification.
- Information Gain evaluates the reduction in entropy after splitting.
- The algorithm selects splits according to the chosen criterion.
- Model complexity should be controlled to improve generalization.

THANK YOU